BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

MATHEMATICS (10th) — (Fresh/Re-appear) — (Version-C) — SECTION – A

Time Allowed: 20 Min | **Maximum Marks:** 15 | **Paper Code:** BN-XXXII-I

Note: Note: There are THREE sections in this paper i.e A, B & C. Attempt section A and return it to the superintendent within the given time. Q.No.1: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

Q1. If R = {(0,0), (8,2), (10,3), (14,12)}, the Dom(R) is

(A) {0,8,10,14}
(B) {0,2,3,12}
(C) {8,10,4}
(D) {0,10}

Q2. Find the mode of the data, 3,2,1,1,1,5,3,1,2,1,2

(A) 1
(B) 2
(C) 3
(D) 4

Q3. $1+\tan^2\theta =$ _____

(A) $\sin^2\theta$
(B) $\cos^2\theta$
(C) $\operatorname{cosec}^2\theta$
(D) $\sec^2\theta$

Q4. The solutions of the equation $x^2 + 7x - 18 = 0$ are

(A) 2 or -9
(B) -2 or 9
(C) -2 or -9
(D) 2 or 9

Q5. If 7:9 :: x:27 then

(A) x = 21
(B) x = 3
(C) x = 7
(D) x = 81

Q6. $(x^2+2) / (x^2+2x+2)$ is

(A) Proper fraction
(B) Improper fraction
(C) Irrational fraction
(D) None of these

Q7. a/b = c/d = e/f = _____

(A) (a-c-e)/(b-d-f)
(B) (a+c+e)/(b-d-f)
(C) (a+c+e)/(b+d+f)
(D) (a-c-e)/(b+d-f)

Q8. (A∪B)' _____

(A) A'∪B'
(B) A'∩B'
(C) A∪B
(D) A∩B

Q9. The number of times a value appears on a set of data is called _____

(A) Frequency
(B) Average
(C) Mode
(D) Median

Q10. A chord is 5cm from the center of a circle of radius 13cm. The length of the chord is;

(A) 6cm
(B) 12cm
(C) 24cm
(D) 30cm

Q11. Sum of roots of $2x^2-3x-4=0$ is _____

(A) -3/2
(B) -2
(C) 4/3
(D) 3/2

Q12. Apollonius is the name of a _____

(A) City
(B) Town
(C) Country
(D) Mathematician

Q13. In a circle, two chords are equally distant from the center of the circle. The chords are

(A) Parallel
(B) Non-parallel
(C) Congruent
(D) Not congruent

Q14. The line which meets the circle at one point is _____

(A) Secant
(B) Diameter
(C) Chord
(D) Tangent

Q15. In the quadratic formula, the expression $b^2 - 4ac$ is called _____ of the quadratic equation $ax^2+bx+c=0$

(A) Discriminant
(B) Roots
(C) Square
(D) Cube

MATHEMATICS (10th) — (Fresh/Re-appear) — SECTION –B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 36

Note: Note: Attempt section B & C accordingly.

Q2. (Marks 36) Q.No 2. Attempt any NINE parts of the following. All parts carry equal marks.

i. Solve the equation : $x^2 - 8x + 15 = 0$

ii. Prove that $(1+2W)(1+2W^2)(1-W-W^2) = 6$

iii. The sum of the squares of three consecutive integers is 50. Find the integers.

iv. Find the mean proportional of 12 and 3.

v. If 8 persons complete a work in 10 days, then how many days would 10 persons take to complete the same work?

vi. Resolve into partial fractions. $(2x+1) / ((x^2+1)(x-1))$

vii. If A = {3,4,5}, B = {5,6,7} and C = {8,9,10} then prove that AU(BUC) = (AUB)UC

viii. If A = {1,2,3,4}, B = {1,3,5} and R = {(x,y) | y < x} is a binary relation from A to B, then wirte it in tabular form.

ix. The following are the weights (in Kg) of students of the 10th grade.
45,30,25,36,42,27,31,43,49 and 50. Calculate mean of the weights.

x. Prove that $(\cos\theta - \sin\theta)^2 = 1 - 2\sin\theta\cos\theta$

xi. Show that $(\sin\theta + \cos\theta)^2 = 1 + 2\sin\theta\cos\theta$

xii. Solve the equation $\sqrt{5x + 21} = x+3$

SECTION –C

Maximum Marks: 24

Note: Note: Attempt any THREE of the following. All questions carry equal marks.

Q3. Prove that : A straight line, drawn from the center of a circle to bisect a chord (which is not a diameter) is perpendicular to the chord.

Q4. Prove that : If a line is drawn perpendicular to a radial segment of a circle at its outer end point, it is tangent to the circle at that point.

Q5. Prove that : The angle in a semi-circle is a right angle.

Q6. Construct a triangle with sides 2cm, 2.5cm and 3cm. Also draw its circumcircle.